

Scalability & Future Plan

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-server deployment	Limited concurrent users	High
2	Static ML model	Needs periodic updates	Medium
3	Basic security	Production readiness limited	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade /Solution
1	User Load	Small-scale	Cloud deployment with load balancing
2	Data Storage	Local files	Cloud database
3	Performance	Basic	Caching and optimization
4	Security	Basic validation	Authentication & HTTPS

Future Roadmap:

Phase 1	Cloud deployment	3 Months	Supports more users
---------	------------------	----------	---------------------

Phase 2	Fertilizer Recommendation, Multi-language Support & Farmer Dashboard	6 Months	Enhances usability and provides complete farming assistance
Phase 3	Real-Time Weather API Integration	9 Months	Improves crop recommendation accuracy
Phase 4	Android Mobile Application	12 Months	Provides easier access for farmers